Serological technique

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DEFINITION

- **Serology**: The scientific study of blood serum and other bodily fluids(plasma, semen, saliva, and cerebrospinal fluid (CSF)) for the identification of antibodies.
- **Antibodies** are typically formed in response to:
- An **infection** (microorganism)
- Against **foreign** proteins (mismatched blood transfusion)
- Against **own** proteins (autoimmune disease).

- **Antibody (Ab):** Also known as an **immunoglobulin** (Ig): It is **Y-shaped** protein produced mainly by plasma cells that is used by the immune system to neutralize the immunogen(pathogen). once an antibody and antigen bind, they become an **immune complex**

- **There are 5 types of antibodies:**

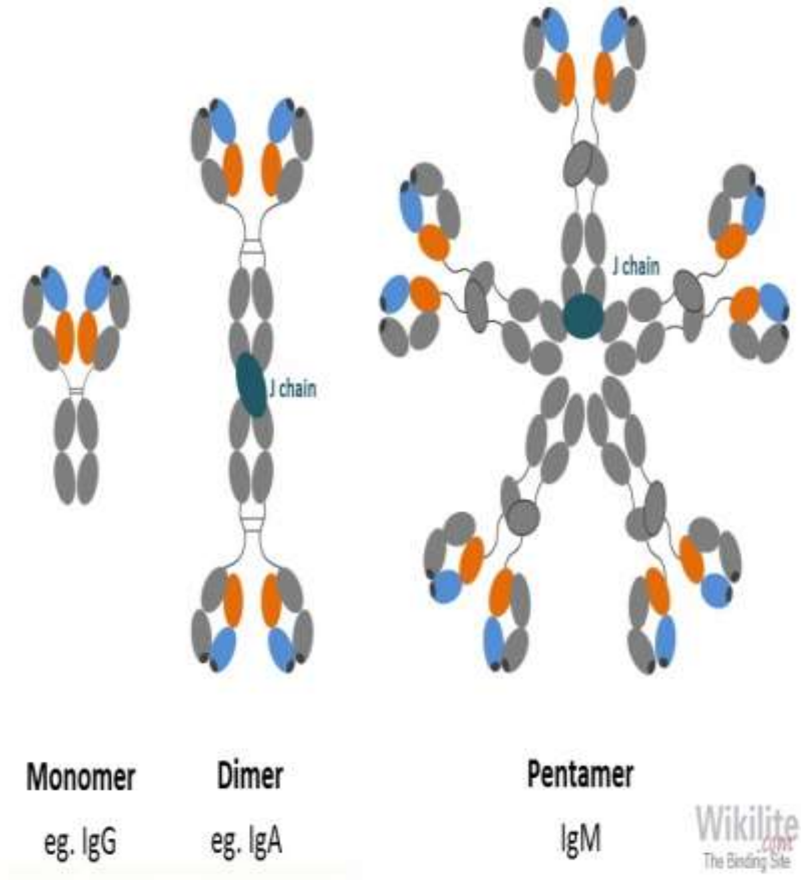
1- IgG: IgG 1, IgG 2, IgG 3, IgG 4

2- IgM

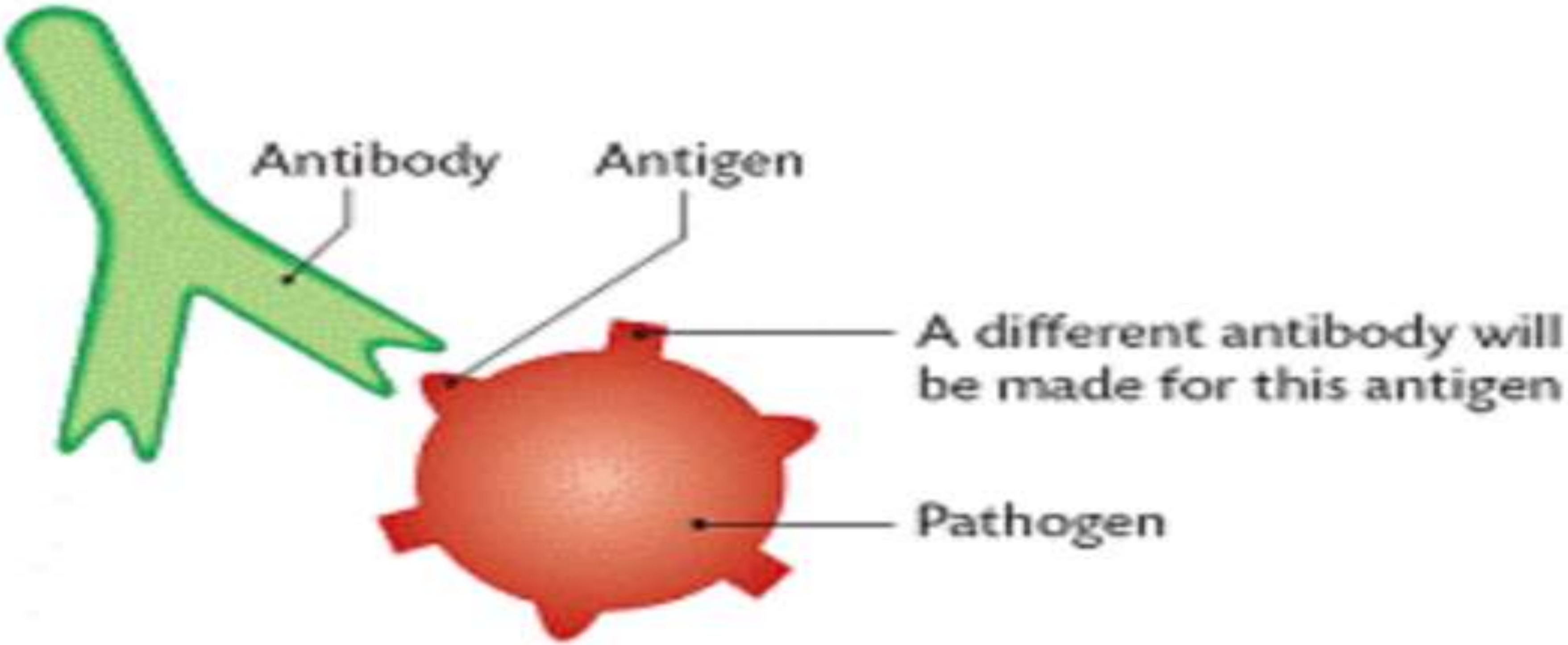
3-IgA: IgA1, IgA2

4- IgE

5- IgD



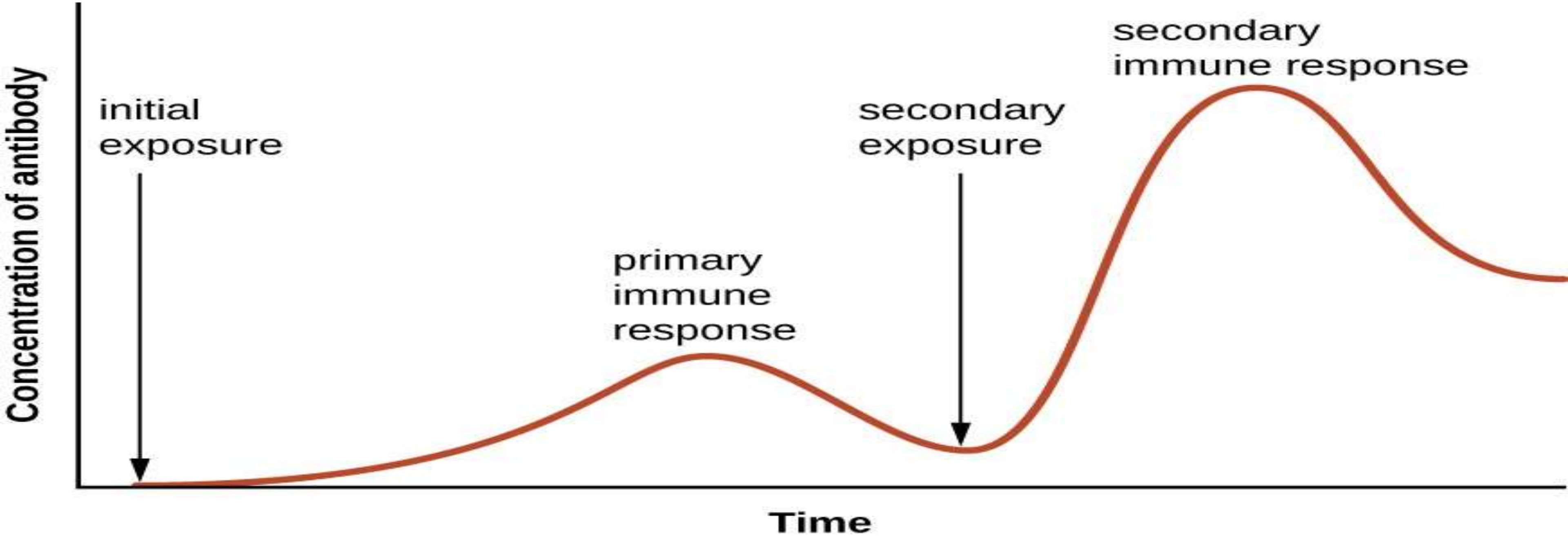
Antigen: Is a substance that **may induce immune response** in the body and react specifically with product of immune response.
Immunogen: Is a substance that **induce immune response** in the body and react specifically with their products.



- **Two types of responses** – according to exposure to antigen

1-Primary immune response (first exposure), takes 2 weeks mediated by IgM.

2-Secondary immune response (second exposure to same antigen) mediated by memory cells that secrets IgG.

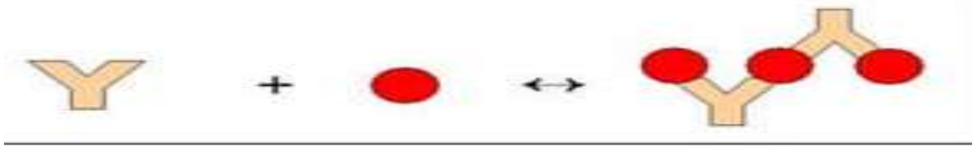


Key Points

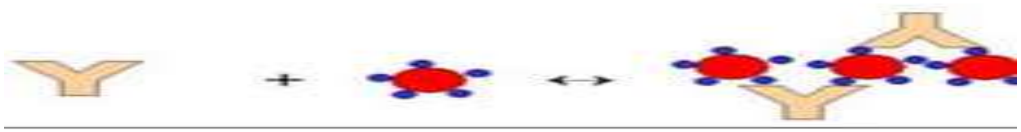
- Serology is based on detecting immunoglobulin levels during the **course of an infection**.
- They can be used for **diagnostic** and **epidemiological** studies.
- Serological techniques can **differentiate** between IgM and IgG antibodies, thus **determining the stage** of the infection.
- **immunoglobulin M**: Largest antibody produced by B cells and the first to appear in response to initial exposure to antigen(A primary immune response).
- **immunoglobulin G** (IgG) : Most abundant antibody isotype , it is highly specific and bind more tightly (A secondary immune response).
- A ratio high in IgM indicates that an infection is in its early stages, while a ratio high in IgG indicates that the infection is in its later stage.

- **Various techniques can be used to detect the presence of antibodies or antigens**
- **Agglutination Tests**
- Simple to perform, require no special
- equipment, and are usually less expensive.
- **Principle of agglutination tests**
- The reaction between a particulate antigen and an antibody results in visible clumping called agglutination .

- **I. Active agglutination slide tests**
- These is a direct agglutination of
- bacterial antigen with its corresponding antibody
- (**slide agglutination of salmonellae)**

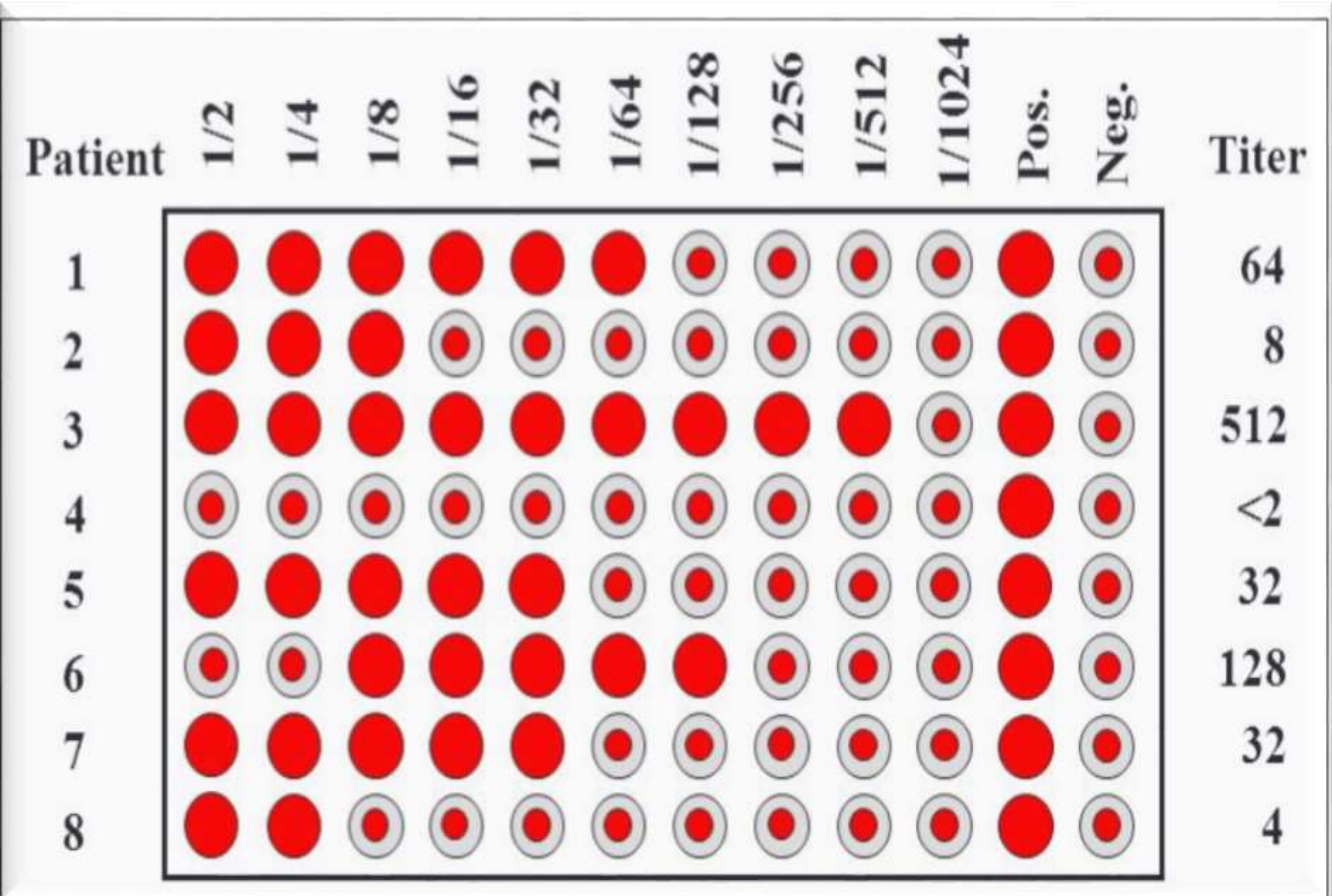


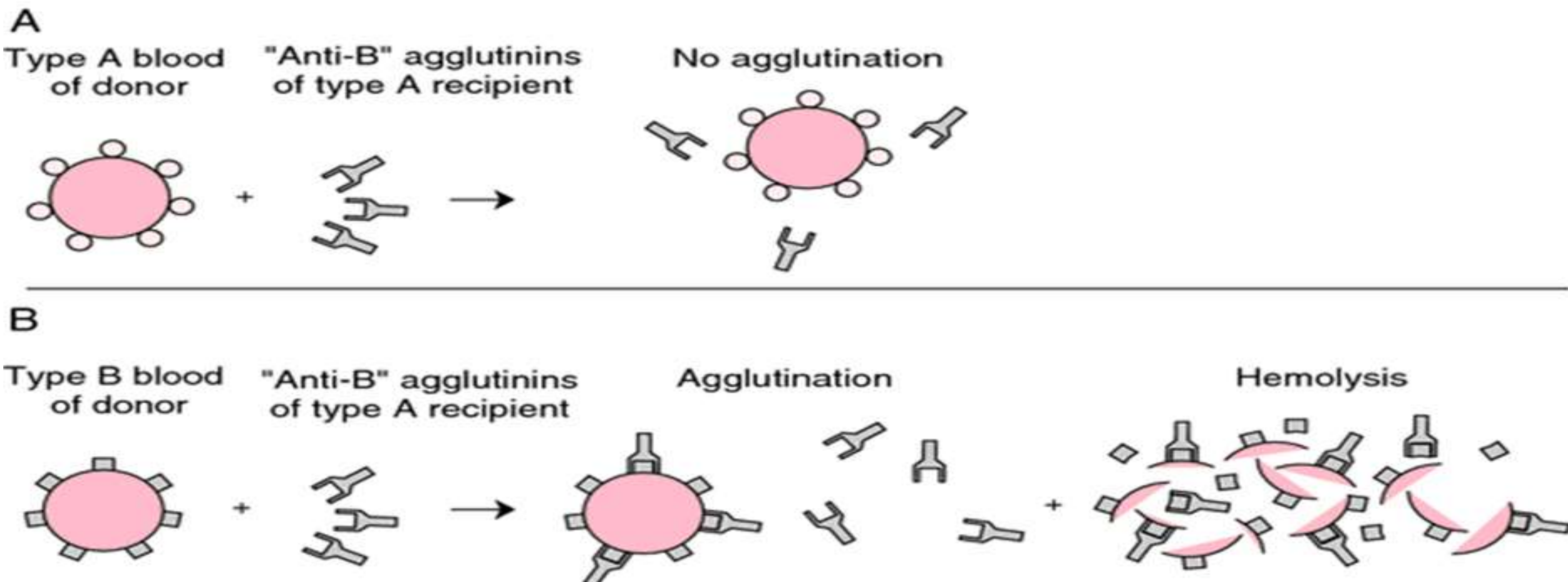
- **II. Passive agglutination slide tests**
- These are tests in which the specific antibody or known antigen is attached to **a particles** or **cells**. Such as **Latex particles** Such as antistreptolysin O (ASO), CRP
- .

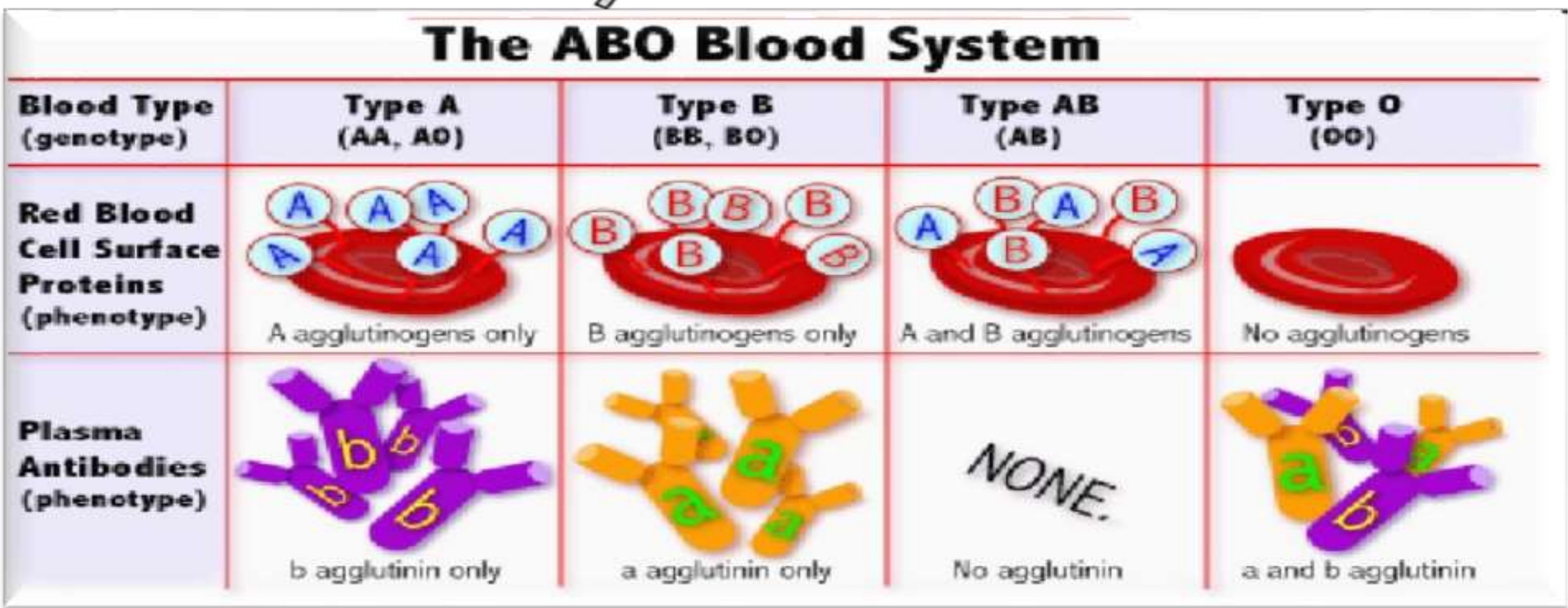


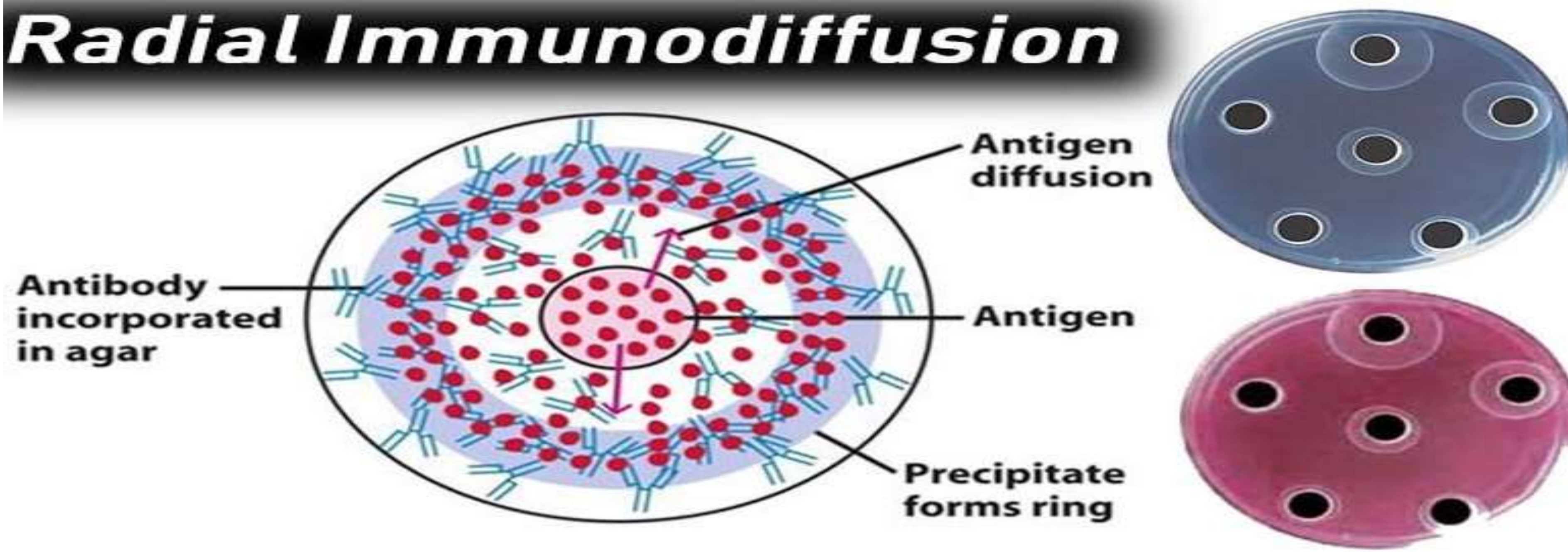
- **Latex particles:**
- these are polystyrene particles that can be coated with either known antigen or specific antibody.

- **Prozone** effect is the **invisibility of agglutination** at high concentrations of antibodies, it forms very minute complexes that do not clump to form visible agglutination.
- **Qualitative agglutination test**
- The antibody is mixed with the particulate antigen and a positive test is indicated by the agglutination of the particulate antigen(blood type)
- **Quantitative agglutination test**
- Serial dilutions of the sample can be made and it is tested for antibody. Then a fixed amount of particulate antigen or bacteria or red blood cells can be added to it.
- Determine the maximum dilution which forms agglutination and this **maximum** dilution which gives observable agglutination is known as the **titer**.


Patient
1/2
1/4
1/8
1/16
1/32
1/64
1/128
1/256
1/512
1/1024
Pos.
Neg.
Titer
1
2
3
4
5
6
7
8
64
8
512
<2
32
128
32
4


A
Type A blood of donor
"Anti-B" agglutinins of type A recipient
No agglutination
B
Type B blood of donor
"Anti-B" agglutinins of type A recipient
Agglutination
Hemolysis


The ABO Blood System
Blood Type (genotype)
Type A (AA, AO)
Type B (BB, BO)
Type AB (AB)
Type O (OO)
Red Blood Cell Surface Proteins (phenotype)
A agglutinogens only
B agglutinogens only
A and B agglutinogens
No agglutinogens
Plasma Antibodies (phenotype)
b agglutinin only
a agglutinin only
NONE.
No agglutinin
a and b agglutinin



Radial immunodiffusion (RID):

Is based on the precipitation of antigen and antibody to an insoluble precipitin complex and thus directly measures the level of antibody(IgG) concentration in serum or plasma. It takes at least 24 h to perform.

Used in detection levels of immunoglobulns and complement component.

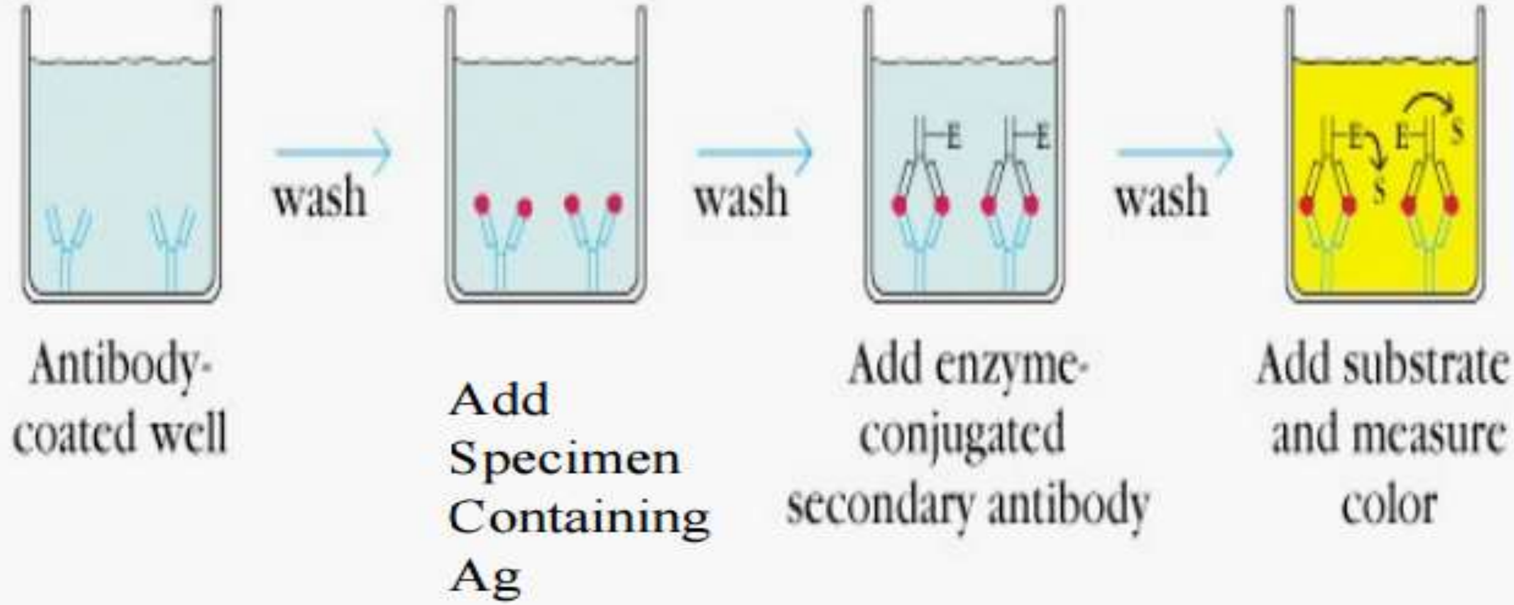
- Antibody incorporated (mixed) into the gel.
- Antigen added to wells.

Enzyme-linked immunosorbent assay (ELISA):

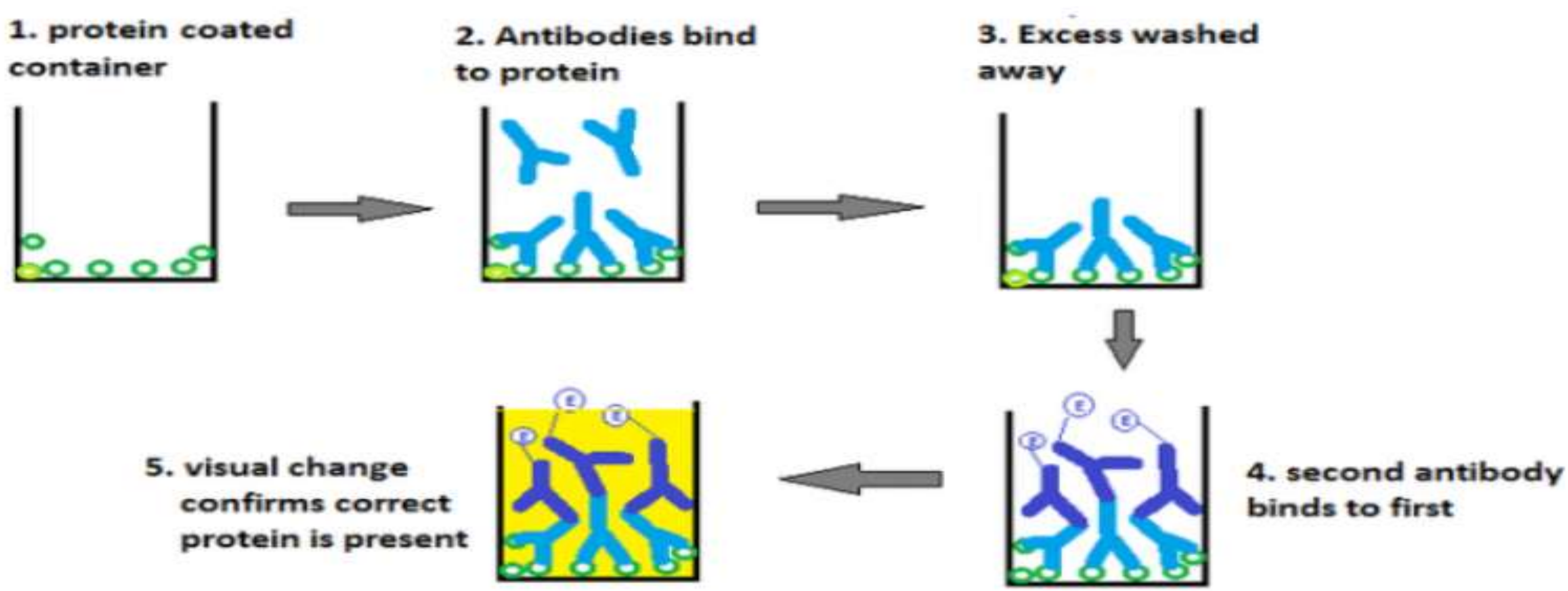
This assay is specific, sensitive based on

antigen-antibody binding reactions (Immune complex).

- An **enzyme** is used as a marker to measure the formation of these **complexes**.
- The technique performed by specialized equipment and trained personnel and require a small amount of specimen.
- The assay uses a blood sample(serum or plasma) and take several hrs.
- ELISA techniques are used in the **diagnosis of microbial infections**.
- The **results of qualitative ELISA techniques** can be read visually.
- **Large numbers** of specimens can be tested at one time.
- **There are two main ways of performing ELISA**
- Double or **direct** antibody technique, to detect antigen
- **Indirect** technique, to detect and assay antibody



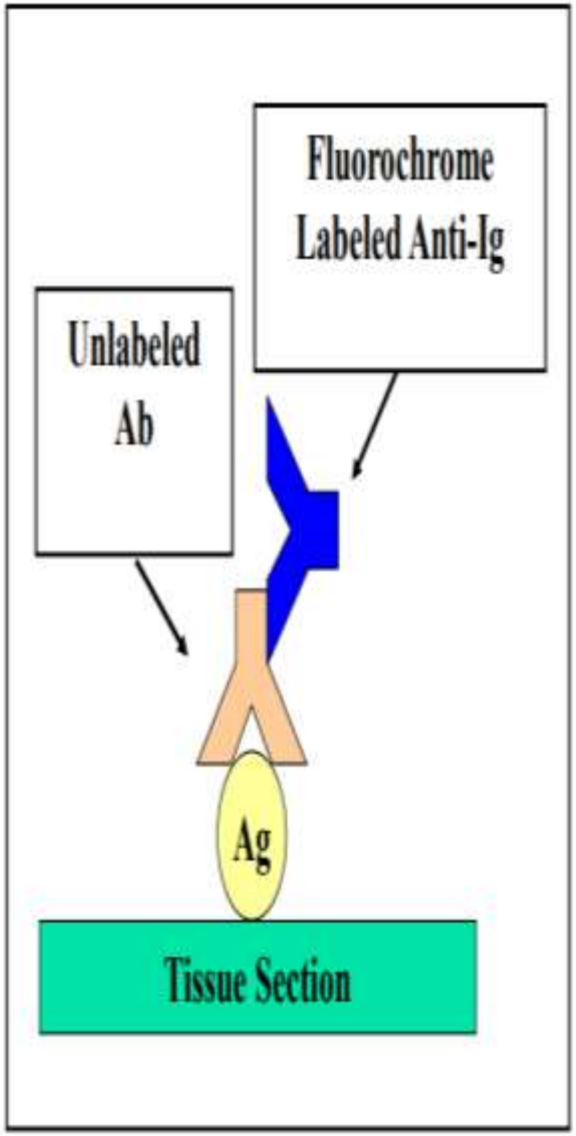
Indirect ELISA



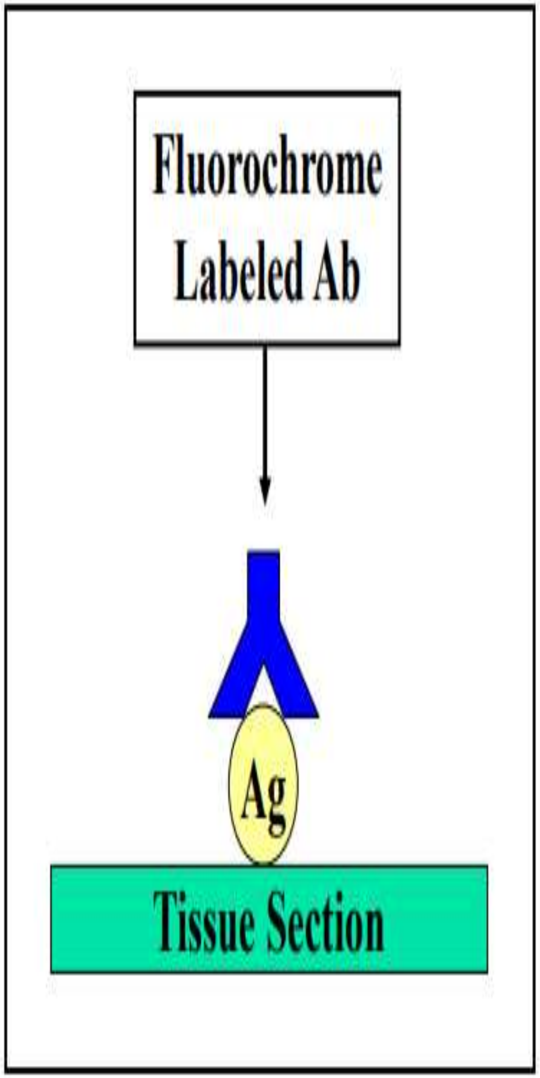
- **Immunofluorescence tests or fluorescent antibody tests (FAT)**:
- This method is rapid and specific.
- **Principle:** identify the presence of **antigens** in infected cells by their ability to react with specific antibodies, which **conjugated** with Fluorescent dyes (fluorochromes) and can be viewed with a **fluorescent microscope**.

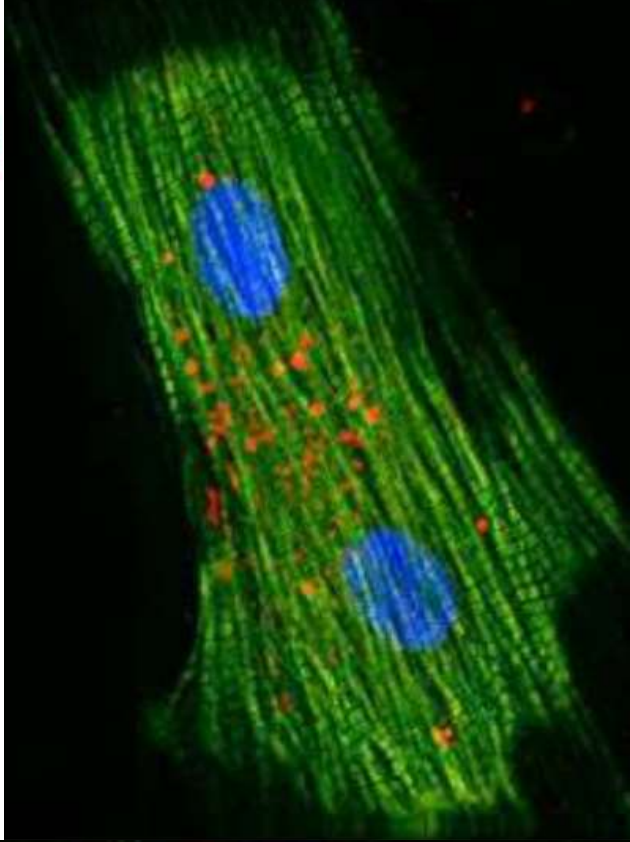
- **A. Direct fluorescent antibody tests**
- Is used to **detect and identify an unknown antigen** in specimens and It is called direct test because the **fluorescent dye** is attached, or **labeled, directly** to the antibody.
- **Direct FAT can be used:**
- To **identify bacteria** when the numbers are very **low**,
- To **detect viruses** growing in **tissue culture** such as **rabies virus** in the brains of infected animals or antigens of **HIV** on the surface of infected cells.
- **B. Indirect Fluorescent antibody test**
- **Unlabeled antibody** combines with **antigen** and the **immune complex** is detected by attaching a **fluorescent-labeled antispecies globulin** to the **antibody**. The antibody, therefore, is labeled **indirectly**.
- Indirect FAT is used in two main ways:
- To **detect and identify unknown antigen** in specimens
- To **detect antibodies** in a patient serum using a **known antigen** (microorganism).

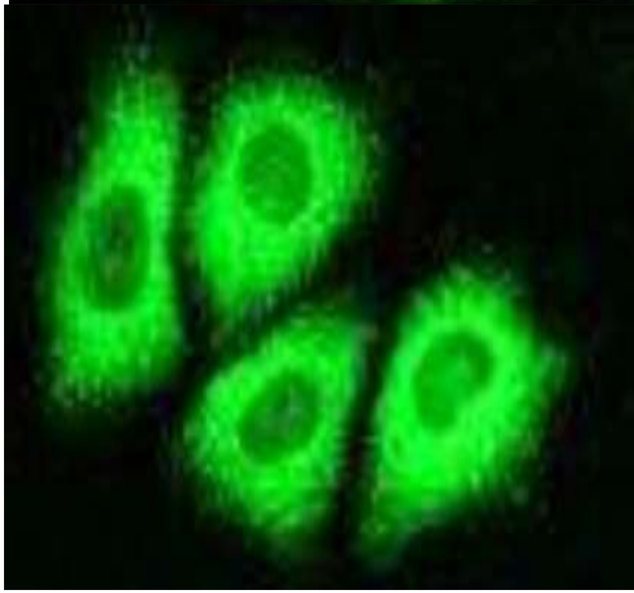
Indirect Immunofluoroscence Test



DIRECT IMMUNOFLUORESCENCE





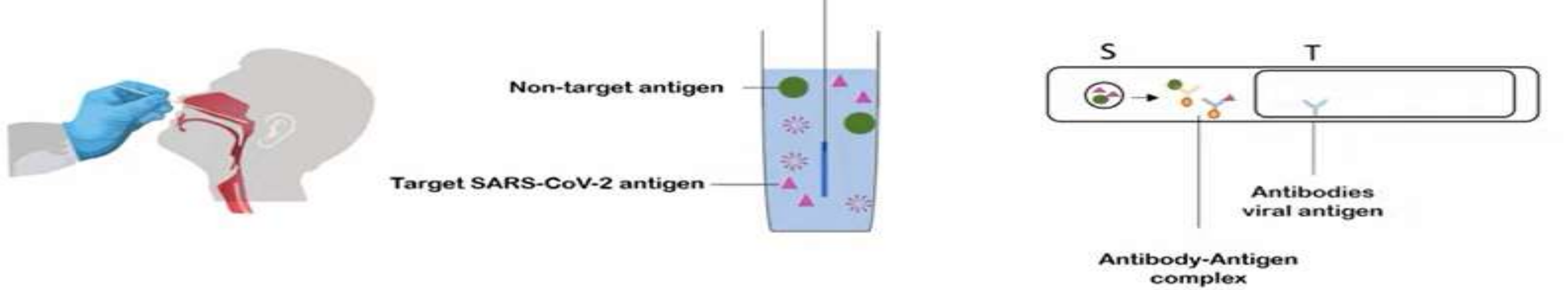


- **A lateral flow test (immunochromatographic assay)**
- It is a rapid diagnostic test that detects the presence or absence of a target analyte (antigen or antibody) in a liquid sample (blood, serum, urine, saliva) as
- Pregnancy test (hCG)
- COVID-19 rapid antigen test
- **Principle of Lateral Flow Test**
- The test works on antigen–antibody specific binding

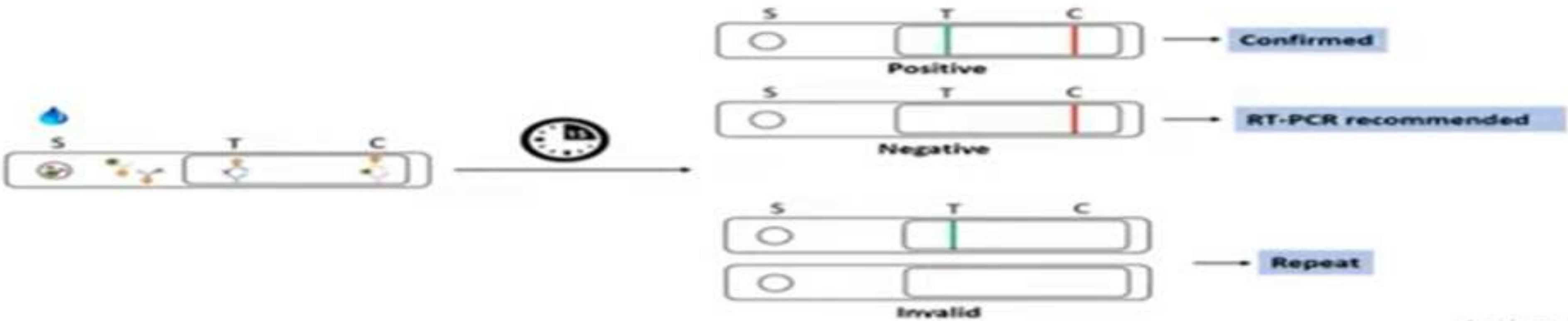
If the target is present in the sample, it binds to labeled antibodies and forms a visible line on the strip.

- **Sample application**
- The liquid sample is added to the sample pad.
- Capillary flow The sample moves laterally along the strip by capillary action (no external force).
- Immune recognition: The target (antigen or antibody) binds to a labeled antibody/antigen (usually gold nanoparticles or latex beads).
- Capture and visualization, the immune complex is captured at the test line •
- A control line confirms proper flow and test validity.

- **Lateral flow tests(immunochromatographic test):**
- **Sample pad**: It acts as a sponge and holds an excess of sample fluid. Once soaked, the fluid migrates to the second element.
  Conjugate pad: A bio-active particles in a salt-sugar matrix that guarantee an optimized chemical reaction between the target molecule (e.g. an antigen) and its chemical partner (e.g., antibody) that has been immobilized on the particle's surface.
- **Control**: It contains an antibody that picks up free latex/gold in order to confirm the test has operated correctly.
- **Test:** It contains a specific capture molecule and only captures those particles onto which an analyte (antigen or antibody) molecule has been immobilized.
- Observe results and interpret accordingly.
- Test negative: Only on the band at the control region
- Test positive: Both bands at test and control regions
- It results in generating a colored red-purple line indicates the presence of antigen of interest in the sample within 15 minutes.



Result Interpretation



	Antigen kit	Antibody kit	RT-PCR kit
Sample	Nasopharyngeal	Blood	Nasopharyngeal
Principle	Detects viral antigen using Monoclonal antibody	Detects antibodies against viral antigen	Detects viral genetic Material using primers fluorescent probes
Test Time	15 minutes	15 minutes	3-5 hours
Sensitivity	Low	Low	High
Infection	Current*	Previous and/or Current*	Current*
Ease	Easy	Easy	Lab facility Trained per